The Unified Theorem of Discrete and Continuous: From Yin-Face Adhesion Dissolving Quantum Gravity Divergence to the Luoshu Matrix Projection

— A Supplementary Note to Volume V4 of The Luoshu Master Equation

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0. Positioning and Annotation Convention

This appendix is a supplementary explanatory appendix to V4, *Modern Geometric Reconstruction of the Luoshu Master Equation: Fiber Bundles, Resonance K, and the Fractal Center Point 5*. This appendix **does not modify any published conclusions of V3/V4**; all experimentally compared results of V3/V4 that have passed numerical verification are retained.

This appendix annotates the argumentative status of each proposition at three levels:

- **【Proven】**: already derived in V3/V4, or a standard physical fact. This appendix reproduces the proof process so that the reader need not consult the original papers.
- **【Hypothesis】**: a physical hypothesis or conjecture proposed in this appendix, awaiting rigorous derivation.
- **【Image correspondence】**: a correspondence between a physical image and a modern physical concept; not claimed to be a rigorous theorem.

Symbols are consistent with V4: $\Phi = (\sqrt{5}-1)/2$ is the golden ratio; ℓ_P is the Planck length; v is the basic spacing of energy- mass particles; n is the base- space layer number; Δn is the layer- number difference; K is the generalized first Chern number (resonance K); FCA is the Fractal Continuity Axiom (V3 Axiom F).

Part One: The Yin- Face Adhesion Argument for

Discretization and Continuization

J.1 Definitions of Yang Face, Yin Face, and Void

Definition J.1: Following V3 Section 1.9:

- **Yang face**: $n \geq 0$, energy- mass particle spacing $\Delta \geq \ell_P$, i.e., the observed world above the Planck scale.
- **Yin face**: $0 < \Delta < \ell_P$, the continuous spectrum in which the energy- mass particle spacing continuously shrinks from the Planck length to the infinitesimal.
- **Void**: $\Delta = 0$, the static- energy block, in which energy is completely at rest and unmanifested, producing no force, no light, and no observable effect whatsoever.

【Proven】(reproducing V3 §1.9.1–1.9.5)

V3 Proposition 1.9.1: The yin face is the continuous spectrum of energy- particle spacing from zero to the Planck length. The proof has five steps: first, the starting point of the yin face is the void, where the spacing equals zero; second, from the void "seeping out," a tiny increment in spacing appears, and the yin face begins; third, the spacing continuously increases from $\Delta_k = \ell_P \times 24^{-k}$ ($k = \infty, \ldots, 0$) to ℓ_P; fourth, the yin face is not "a point" but a continuous spectrum; fifth, **the void is the "cold reservoir" of energy**, and energy particles come from the void.

V3 Proposition 1.9.2: The yin face has a hierarchical structure. The hierarchy is defined by the magnitude of the spacing, $\Delta_k = \ell_P \times 24^{-k}$; the spacing of each layer is $1/24$ of that of the previous layer, symmetric with the yang face's 24- fold amplification per layer. (V4 Definition 1.3.1 defines the base space as the layer- number axis, $v_n = v \times 24^n$.)

V3 Proposition 1.9.3: The Planck scale is the boundary layer between yin and yang. The yin face is $0 < \Delta < \ell_P$, the yang face is $\Delta \geq \ell_P$, and ℓ_P is the boundary; energy particles surge out of the void, pass through the Planck scale, and enter the yang face—this is "yin- face upwelling."

V3 Proposition 1.9.4: The three sets of velocities form a continuous spectrum. The yang- face velocity is fixed at the speed of light c; the yin- face velocity is not a single value but **a continuous spectrum continuously increasing from the speed of light c to infinity**—as $\Delta \to \ell_P$, $v_{\text{yin}} \to c$; as $\Delta \to 0$, $v_{\text{yin}} \to \infty$; at the void $\Delta = 0$, $v = 0$. The three sets of velocities together: void $v = 0$, yang face $v = c$, yin face $v \in (c, \infty)$ continuous spectrum.

V3 Proposition 1.9.5: The void and Center Point 5 are two perspectives on the same thing. Seen from outside, Center Point 5 is the central node of the Luoshu; seen from inside, Center Point 5 is the void. The umbilical cord is the soft skin of the spring itself—it is the layer of connective channel that can connect everywhere; the spring skin starts from Center Point 5 and extends layer by layer outward, connecting all layers into one body. Center Point 5 is the void, and the void connects all.

A supplementary sentence in this appendix: The void is not "nothing"; it is

undifferentiated energy and the organizational center (V3 §1.8.2, §1.9.5). This is a reproduction of the original V3 text, not a newly added hypothesis.

Umbilical cord and connection: V4 Section 1.10 defines the connection $A = A_n dn$, with connection coefficient $A_n = \ln(1/\Phi^5) \approx 2.406$. The umbilical cord is the physical carrier of this connection—it is the soft skin of the spring, not a tail; it starts from Center Point 5, extends along each layer, and connects different layers. The longitudinal umbilical channel retains the Φ^5 attenuation; the transverse closed channel corresponding to gravity does not use this attenuation (V4 Chapter 7). Therefore "umbilical cord" is not "tail"; it is **the soft skin/connective channel that connects everywhere**.

Proposition J.1 (Single‑Layer Discreteness Leads to Divergence)

Proposition: If space is treated as a single discrete grid and the energy density or curvature is calculated directly at the grid fracture, divergence appears at the fracture. Let the grid spacing of that layer be v_n; under the continuous‑field‑theory approximation, the local energy density at the fracture is $\rho \propto 1/v_n^2$; as $v_n \to 0$, $\rho \to \infty$.

Proof sketch (elementary dimensional analysis):

Let the single‑layer grid spacing be v_n. The volume element at the fracture is $\sim v_n^3$; if the energy is finite, the energy density is $\rho \sim E/v_n^3$; if only a one‑dimensional fracture is considered, the linear density is $\sim E/v_n$; for a two‑dimensional cross‑section, $\rho \sim E/v_n^2$. In any case, as $v_n \to 0$, divergence occurs. This is the geometric origin of "ultraviolet divergence" in standard field theory: treating a certain layer of the grid as the only layer and forcibly calculating continuous field quantities at the fracture.

The perspective of this theory: The single‑layer discrete image takes only one layer of the fractal nesting and ignores the filling of the fracture by the deeper $(n-1)$‑layer grid. V4 Chapter 11 has proven: quantum mechanics is the coarse‑graining of a certain layer of the grid, and general relativity is the smooth limit of the superposition of multiple grid layers; the root of their conflict is precisely that each treats "a certain layer" as "the only layer." (V4 Proposition 11.4.1: The quantum gravity problem resolves naturally—quantum mechanics is a small‑scale coarse‑graining approximation, general relativity is a large‑scale smooth approximation, and the two are projections of the same fractal fiber bundle under different limits, naturally failing at intermediate scales.)

【Standard physical fact; this appendix uses it as an analogy and does not claim that this theory has proven divergence】

Proposition J.2 (Fractal Nesting + Yin‑Face Adhesion

Dissolves Divergence)

Proposition: The space of this theory is a fractal grid nested layer by layer. When approaching a single‑layer fracture, the yang face dissolves the singularity of the "bare fracture," and the yin face dissolves the operation of "single‑layer truncation" itself. The two have different roles and must not be conflated.

Yang‑face dissolution (reproducing the fractal‑nesting proof of V4 Chapter 1 and Chapter 11)

V4 Chapter 1 defines the base space $B = \{n \in \mathbb{R} | 0 \le n \le n_{max}\}$, with 24‑fold amplification per layer, and the n‑th layer grid spacing $v_n = v \times 24^n$. Between two adjacent grid points of the n‑th layer, there exists a complete grid unit of the $(n-1)$‑th layer (because 24‑fold amplification means each layer contains 24 sub‑units). Therefore, the "fracture" of the n‑th layer is, from the perspective of the $(n-1)$‑th layer, merely an ordinary grid interval, with no singularity. Pushing upward layer by layer, the fracture at any scale is filled by a deeper grid—this is the dissolution of the singularity of the "bare fracture" by fractal self‑similarity.

But the yang face itself remains discrete: under single‑layer truncation, divergence still appears. What the yang face dissolves is the singularity of the "bare fracture," not divergence itself. This point must be stated clearly; otherwise it will be read as "the yang face has already eliminated divergence."

Yin‑face dissolution (reproducing V3 §12.3.7 "same K value" and FCA Axiom F)

V3 Proposition 12.3.7: Center Point 5 is the void. The center of the Luoshu is 5; 5 connects the 8 peripheral nodes; 5 itself is empty; it is the intersection of all holes and the center of the sphere in three‑dimensional space.

V3 Proposition 12.3.8: Twin primes share the same K. The digital roots of the twin primes $(9K+2, 9K+4)$ are 2 and 4 respectively; both contain the common term $9K$, so they share the same K. K is the "number of turns" (depth layer); sharing K means they are located at the same "depth layer" in the layer‑number coordinate, connected through a connective structure, just as two atoms are connected through a chemical bond.

Proof essentials (V3 §12.3.1–12.3.2): Let P and $P+2$ be twin primes, both greater than 3. By the prime digital‑root constraint, primes greater than 3 can only fall on the six stations of the times‑2 orbit $\{1,2,4,5,7,8\}$. Enumerating digital‑root pairs: $(1,3)$ not present, $(2,4)$ present, $(4,6)$ not present, $(5,7)$ present, $(7,9)$ not present, $(8,1)$ present. Therefore the digital‑root pairs of twin primes can only be $(2,4)$, $(5,7)$, $(8,1)$. The three types correspond to K values: Type I: $9K+2$ and $9K+4$, e.g., 11 and 13; Type II: $9K+5$ and $9K+7$, e.g., 5 and 7; Type III: $9K+8$ and $9K+1$ $(= 9(K+1)+2)$, e.g., 17 and 19. The three types of twin primes share the same K value.

V3 Proposition 12.3.9: K is the same in the void below Center Point 5. Seen from outside, K is a different depth layer; seen from inside, K is all the same void. Twin

primes share K because they are the same in the void below Center Point 5.

V3 Axiom F (FCA, Fractal Continuity Axiom): On the yin face (layer spacing $\to 0$), the interval between discrete integer grid points is not a "gap" but a connective structure filled by the spiral trajectories of energy particles; as the layer spacing $\to 0$, the length of the connective structure tends to zero, and the discrete grid points become a continuous tiling in the layer‑number coordinate. FCA asserts: when the fractal distance tends to zero, discrete grid points "adhere" into the same node in the fractal limit.

Non‑standard‑analysis formalization of FCA (V3 §12.3.11): In the hyperreal field $({}^*\mathbb{R})$, a fractal‑node equivalence relation $\sim_F$ is introduced. FCA‑NSA asserts that this equivalence relation is not trivial on infinite elements. If ZFC is consistent, then ZFC+FCA‑NSA is consistent (V3 Proposition 12.3.11.1); that is, FCA is neither provable nor refutable in standard number theory and is an independent axiom.

The key to yin‑face dissolution

As $\Delta n \to 0$, layers adhere to one another; the continuous spectrum of the yin‑face velocity increasing from c to infinity degenerates at the limit into "no distance to transmit," and there is no longer the operation of "separating out a certain layer alone." What the yin face dissolves is the operation of "single‑layer truncation" itself, not "calculating divergence as finite." In V3's words: the discrete jumps of the yang face (Cantor‑type gaps) and the continuous adhesion of the yin face (same K value) are two sides of the same fractal structure at different scales.

Conclusion: Divergence is not physical reality but a mathematical artifact of "single‑layer truncation"; yin‑face adhesion is the topological mechanism that dissolves the "truncation operation."

【Argument of this appendix; the "adhesion" part relies on V3 Axiom F (FCA) and holds within the FCA axiom system; it does not claim to have proven within standard ZFC that infinity does not exist】

Proposition J.3 (Continuity and Discreteness Are Different Layers of the Same Fractal Structure)

Proposition: What quantum mechanics sees is the discreteness of a certain layer of the grid; what general relativity sees is the continuous limit after multiple grid layers are superimposed. The two are projections of the same fractal fiber bundle at different layers, not two mutually exclusive ontologies.

【Proven】(reproducing V4 Chapter 11)

V4 Proposition 11.2.1: The Schrödinger equation is the approximation of the Luoshu master equation in the limit of no driving, no coupling, and no dissipation. The Luoshu master equation is $i\hbar\partial\psi/\partial t = \hat{H}_{\text{Luoshu}}\psi$, where

$\hat{H}_{\text{Luoshu}} = \hat{H}_0 + \hat{H}_{\text{drive}} + \hat{H}_{\text{coup}} - i\hbar\hat{\Gamma}$; setting the driving term, coupling term, and dissipation term to zero gives

$i\hbar\partial\psi/\partial t = [\hbar^2\hat{k}^2/(2m) + V_0\hat{\Phi}^n]\psi$; when the exponential potential approximates the Coulomb potential (V4 Proposition 4.5.2), it reduces to the standard Schrödinger equation.

V4 Proposition 11.3.1: The Einstein field equation is the emergence of the local curvature average of the fiber bundle under large- scale smoothing. General relativity is the effective description of the gravitational closed channel in the smooth large- scale limit; the curvature tensor corresponds to the local curvature average of the fractal fiber bundle at large scales, and the energy- momentum tensor corresponds to the effective mass source produced by manifold intersections.

V4 Proposition 11.4.1: The quantum gravity problem resolves naturally. Quantum mechanics is a small- scale coarse- graining approximation, general relativity is a large- scale smooth approximation, and the two are projections of the same fractal fiber bundle under different limits, naturally failing at intermediate scales—because there neither coarse- graining nor smoothing can be performed.

V4 correspondence table (V4 §11.5):

Modern physics concept	Corresponding object in this theory	Approximation condition
Wave function	Fiber phase field	Coarse- grained statistical average
Planck constant	Quantization unit of the finest step of the fractal grid	Basic constant of the discrete grid
Speed of light c	Topological speed of the spiral cross- section advancing along the fiber	Determined by the fiber connection
Lorentz transformation	Grid projection scaling when the observer moves relatively	Large- scale smoothing
Spacetime curvature	Local curvature average of the fiber bundle	Large- scale smoothing
Quantum entanglement	Shared topological connection at the fiber level	Base- space distance large, fiber- space distance small
Uncertainty principle	Geometric lower bound of the discrete fractal grid	Grid cannot be further divided

Wave- particle duality	Particle nature of the vortex standing- wave knot + wave nature of spiral propagation	Two sides of the same vortex

A supplementary sentence in this appendix: V4 Chapter 11 proves the isomorphism of continuity and discreteness from the side of "limit correspondence"; this appendix completes it from the side of "yin- face adhesion"—the "opposition" between continuity and discreteness arises because we work only on the yang face ($\Delta \geq \ell_P$); once we enter the yin face ($0 < \Delta < \ell_P$), discrete grid points form a continuous bridge through topological adhesion, and the unity of discrete and continuous is no longer a philosophical assertion but a topological fact.

【Reproduction within the proven framework; this proposition adds no new physical hypothesis】

Proposition J.3bis (The Irreconcilability of Quantum Mechanics and Relativity Within Their Respective Domains of Applicability)

Proposition: Quantum mechanics (QM) and general relativity (GR), as two independent theoretical systems, possess a fundamental irreconcilability within the domains each claims to apply to. This irreconcilability is not a computational- technical problem but an ontological conflict, manifested specifically at three levels:

Contradiction One: Irreconcilability of state- space structure

The basic assumption of QM is: the state space of a physical system is a Hilbert space H (a complex linear inner- product space); the evolution of the state vector $|\psi\rangle$ is described by the Schrödinger equation $i\hbar\partial\psi/\partial t = \hat{H}\psi$, which is linear evolution. The measurement process assumes that the state vector collapses to some eigenstate, which is a nonlinear, non- unitary process.

The basic assumption of GR is: physical spacetime is a four- dimensional Lorentzian manifold $(M, g_{\mu\nu})$; matter and energy determine spacetime geometry through the Einstein field equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, and geometry determines matter motion; this is a fully deterministic nonlinear theory.

Within the fractal fiber bundle framework (V4 Chapter 1): QM sees the discrete projection of a certain layer of the grid (the coarse- graining limit of V4 Proposition 11.2.1), and its Hilbert- space structure originates from quantization on that discrete grid layer; GR sees the continuous limit after multiple grid layers are superimposed (the large- scale smoothing of V4 Proposition 11.3.1), and its Lorentzian- manifold structure originates from the smooth approximation of the fractal fiber bundle at large scales.

Key argument: If QM and GR are limit manifestations of the same theory, their state spaces should be embeddable into a larger mathematical structure. But in the fractal framework, the state space of QM corresponds to the function space $L^2(\mathrm{grid}_n)$ on the discrete grid of a certain layer n, while the state space of GR corresponds to the smooth manifold space under the full-scale limit—these two spaces are mathematically different types of objects (discrete vs. continuous, finite-dimensional truncation vs. infinite-dimensional smoothness), and there is no natural embedding map that makes them consistent at any intermediate scale.

Contradiction Two: Incompatibility of causal structure

QM allows nonlocal correlations (EPR entanglement), i.e., two spatially separated particles can have correlated measurement results, and this correlation does not depend on any signal transmission (Bell-inequality violations have been experimentally confirmed). In standard QM, this correlation is "instantaneous"—measuring one particle immediately determines the state of the other, although no information is transmitted.

GR requires all physical influences to propagate along timelike or lightlike curves and does not allow superluminal causal influences. The causal structure of spacetime is determined by the light-cone structure of the metric $g_{\mu\nu}$.

Within the fractal framework (V3 §12.3, V4 Chapter 11): the geometric origin of quantum entanglement is the topological connection in the yin face—two entangled particles are very far apart in the yang-face base space, but are connected through a connective structure in the yin face ($\Delta n \to 0$) (V3 Proposition 12.3.9; the yin-face velocity continuous spectrum degenerates at the limit into "no distance to transmit," so transmission requires no time). However, this yin-face connection manifests in the effective description of the yang face as "nonlocal correlation," while GR's causal structure only recognizes timelike/lightlike curves in yang-face spacetime.

Key argument: When considering a scenario that requires both quantum superposition and strong gravity (such as Hawking radiation in the black-hole information paradox), QM's unitary evolution requires information conservation, while GR's event-horizon structure seems to lead to information loss. In the fractal framework, this corresponds to: at intermediate scales (scales that can neither be coarse-grained into a discrete grid nor smoothed into a continuous manifold), an irreconcilable tension appears between the yin-face connective structure and the yang-face causal structure.

Contradiction Three: Ontological conflict of the measurement problem

QM's measurement problem: why does measurement lead to wave-function collapse? Is collapse a physical process or an epistemological update? Standard QM has no consistent answer (the Copenhagen interpretation, the many-worlds interpretation, decoherence theory, etc., each have difficulties).

GR has no concept of "measurement"—spacetime geometry is objective reality and does not depend on the observer.

In the fractal framework (V4 §11.2.2 wave-particle duality): the wave function is the

form of the spiral bundle in the continuous limit, and measurement behavior corresponds to the fiber‑bundle cross‑section coupling between the observer and the system. But "collapse" in the fractal image corresponds to: the observer's detection process selects a specific layer n for coarse‑grained observation, thereby "freezing" the discrete projection of that layer. This selection does not occur in the yin face—in the yin face all layers exist simultaneously and are connected through adhesion.

Key argument: QM's measurement problem essentially originates from treating the discrete projection of a certain layer n as "all of reality." GR has no measurement problem because it automatically averages discreteness at macroscopic scales. But when attempting to apply QM to the gravitational field itself (quantum gravity), one must simultaneously handle discrete‑layer projections and continuous geometry, and at this point the ontological conflict between the two frameworks is fully exposed.

Conclusion: The irreconcilability of QM and GR is not a problem that can be solved by "more precise calculation," but an ontological conflict of two effective theories outside their respective domains of applicability. This theory (the fractal fiber bundle) does not reconcile QM and GR; rather, it points out that each is only a projection of the same fractal structure under different limits—their "irreconcilability" precisely proves the need for a deeper unified framework.

【Reproduction within the proven framework + argument of this appendix; the contradiction analysis is based on the V4 Chapter 11 correspondence and V3 yin‑face theory】

J.1bis The Main Ideas of Twin Primes and the Goldbach Conjecture (anchored to $\Delta n \to 0$)

Twin primes: V3 Proposition 12.3.8 has proven that twin primes $(9K+2, 9K+4)$ share the same K. Proof essentials: let P and $P+2$ be twin primes greater than 3; by the prime digital‑root constraint, primes greater than 3 can only fall on the six stations of the times‑2 orbit $\{1,2,4,5,7,8\}$; enumerating digital‑root pairs, only $(2,4)$, $(5,7)$, $(8,1)$ are possible, and all three contain the common term $9K$, so they share the same K.

Main idea ($\Delta n \to 0$): On the yang face, two twin primes are two separated points on the number axis; but on the yin face ($\Delta n \to 0$), the layer spacing tends to zero, and the layer‑number coordinates of the two primes "adhere" into the same K through the connective structure. This is the geometric meaning of V3 Proposition 12.3.9, "K is the same in the void below Center Point 5": seen from outside, they are two primes; seen from inside, they are the same void.

Goldbach conjecture: V3 Proposition 12.3.12 (within the FCA axiom system) gives a conditional proof. Proof essentials: let N be an even number; $N \bmod 9$ has five possibilities; for each residue, there exists a digital‑root pair (r_1, r_2) such that $r_1 + r_2 \equiv N \pmod 9$; by the fractal collision theorem (V3 Proposition 12.3.3/12.3.10, within

FCA), the fractal envelopes of two digital‑root orbits intersect in an infinite set; in the yin‑face limit, the intersection points adhere into yang‑face integers, corresponding to two primes P_1, P_2 such that $P_1 + P_2 = N$.

Main idea ($\Delta n \to 0$): The difficulty of the Goldbach conjecture lies in the bridge from "dense" to "full coverage"; V3's FCA axiom builds this bridge precisely at the yin‑face limit ($\Delta n \to 0$)—discrete grid points adhere at the limit, and a set that was originally "dense but possibly not covering" covers all possible values after adhesion. V3 explicitly states: this is a conditional theorem within the FCA axiom system and does not claim to have proven the Goldbach conjecture within ZFC.

FCA‑NSA consistency (V3 §12.3.11): Through non‑standard analysis, a fractal‑node equivalence relation $\sim_F$ is defined in the hyperreal field $({}^*\mathbb{R})$, and it is proven that if ZFC is consistent, then ZFC+FCA‑NSA is consistent. FCA does not imply the twin‑prime conjecture in standard number theory, nor vice versa; the two are independent propositions.

This appendix emphasizes: The main ideas of the twin‑prime and Goldbach conjectures are both "yang‑face discreteness, yin‑face adhesion"; $\Delta n \to 0$ is the limiting condition under which adhesion occurs, and it is also the key to these two number‑theoretic problems obtaining conditional proofs within this framework.

【Proven (V3 §12.3.7–12.3.12, within FCA); does not claim proof within ZFC】

Part Two: The Spherical Diagram of Three Orthogonal Rings and the Luoshu Matrix Correspondence

Proposition J.4 (Three‑Ring Orthogonal Spherical Closure)

Proposition: Three rings each start from Center Point 5, wind along a set of mutually orthogonal directions, with radius amplified as $r_n = \ell_P \cdot 24^n/(2\pi)$ (V4 §1.4), and after reaching the theoretical cutoff layer of that ring, fold back from the reverse side to Center Point 5, forming three mutually orthogonal closed spherical/horn surfaces; the three spherical surfaces intersect and project onto the discrete grid as 8 peripheral nodes plus Center Point 5.

【Proven】(reproducing V4 §1.4, §1.6)

V4 Definition 1.4.1: fiber $F = S^1$; the n‑th layer spiral characteristic radius is $r_n = \ell_P \cdot 24^n/(2\pi)$.

V4 Definition 1.6.1: cross‑section $s: B \to E$ satisfies $\pi \circ s = \mathrm{id}_B$; spiral motion is the fiber‑bundle cross‑section; intra‑layer step number $s = 5n$, $\theta(s) = \theta_0 + 2\pi s/5$; written by layer number, $\theta(n) = \theta_0 + 2\pi n$.

V4 Chapter 1 states clearly: all spiral cross‑sections start from $n = 0$, i.e., all spirals

surge outward from Center Point 5.

Folding‑back mechanism (reproducing V3 breathing axiom): V3 Axiom D (breathing axiom): Center Point 5 alternates between an expansion phase and a contraction phase; the expansion phase ejects outward, and the contraction phase contracts inward. Expansion (exhalation) corresponds to outward unfolding, and contraction (inhalation) corresponds to inward recovery. After the three rings reach the theoretical cutoff layer, they fold back to Center Point 5 through the contraction phase.

Limitation of the "maximum layer": Here the "maximum layer" refers to the theoretical cutoff layer of that ring (a cutoff determined by the fractal scale and the age of the universe), not the cosmic topological boundary n_{max} of V4 Chapter 12; the latter is the global holonomy folding‑back, while the former is the closure of a single‑ring envelope surface.

Tube projection vs. spherical envelope surface: The main text of V4 Chapter 5 gives the **Tube projections** of the three sets of orthogonal rings (corner ring $K = 10$, edge‑midpoint ring $K = 5$, gravitational self‑ring $K = 5$); the **spherical envelope surface** of this proposition is a continuous rendering of the same structure. The two do not conflict: the pipeline is the discrete projection, and the spherical surface is the continuous envelope; the Luoshu nine‑grid is the two‑dimensional discrete projection common to both.

【Image correspondence; based on V4 §1.4, §1.6 and the V3 breathing axiom; not a newly added physical hypothesis】

Proposition J.5 (The Luoshu Matrix Is a Projection, Not the Ontology)

Proposition: The planar arrangement of the Luoshu nine‑grid

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4 9 2
3 5 7
8 1 6
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is the projection of the three‑ring orthogonal closed structure onto the two‑dimensional grid: the corner ring $2 \to 4 \to 8 \to 6 \to 2$ corresponds to the strong‑interaction channel $K = 10$; the edge‑midpoint ring $1 \to 3 \to 9 \to 7 \to 1$ corresponds to the electromagnetic channel $K = 5$ (vertical and horizontal stacking); Center 5 is the projection of the global topological origin $n = 0$.

【Proven】(reproducing V4 §5.2.1, §5.1)

V4 §5.2.1: the corner ring goes from Center 5 to the edge midpoint in one turn (equivalent step count $1 \times 5 = 5$), and from Center 5 to the corner in two turns (equivalent step count $2 \times 5 = 10$); therefore $K_{\text{corner}} = 10$, $W[10] = 1/\Phi^{10} \approx 122.991$,

and the total weight of 4 pipelines is 491.964.

V4 §5.1: the edge- midpoint ring closes in a single- phase cycle, with equivalent step count $1 \times 5 = 5$, $K_{\text{edge}} = 5$, $W[5] = 1/\Phi^5 \approx 11.090$, and the total weight of 4 pipelines is 44.360; the vertical and horizontal edge- midpoint rings project onto the same set of nodes $\{1,3,9,7\}$, undergoing topological stacking, but they belong to two sets of orthogonal vortex tubes and are physically independent ("stacking is not coincidence," V4 §5.1).

V4 §5.3 light matrix: a 9×9 Hermitian matrix; the 9 eigenvalues are assigned by multiplicity to the four types of interaction: gravitational mode multiplicity 1, strong- force mode multiplicity 4, electromagnetic mode multiplicity 2, weak- force mode multiplicity 2.

Visual effect of the two- dimensional arrangement: The two- dimensional arrangement creates the visual impression that "2486 is placed diagonally"; in three dimensions the three rings are symmetric and orthogonal, with no privileged direction; "diagonal placement" is a visual effect of the two- dimensional projection, not that the ring itself is diagonal.

【Image correspondence; consistent with the topological stacking conclusion of V4 §5, and a geometric- image completion thereof】

Proposition J.6 (The Observer's Position Selects the Facing Direction)

Proposition: In three dimensions the three rings are symmetric, and there is no natural "facing" direction. The observer's position selects a reference orientation; the corner projection in that orientation appears as the four- corner structure 2486, while the other two directions degenerate into the edge- midpoint projection 1397. Without the observer's position, the corner ring in any direction could be called 2486.

【Hypothesis; image explanation, not a proven theorem of V4】

Explanation: This proposition is an image explanation; it does not involve wave- function collapse, nor does it claim that the observer creates geometry. It is formally similar to "measurement selects the basis" in quantum mechanics, but in this theory the observer's selection is classical and geometric.

Part Three: v and Dimension, Five Bands and Five Energy States

Proposition J.7 (v Is the Ruler of Dimension)

Proposition: v is not an absolute constant but the basic spacing of energy- mass

particles of a certain dimensional- state universe. For our universe, $v = \ell_P$; for other dimensional- state universes, $v = v'$. Different v correspond to different cosmic- web grid spacings, and their energy systems are mutually isolated. The 24- fold is the cross- layer scaling rule; v is the reference spacing of each dimensional- state universe itself. The 24 governs "how much to amplify within a layer," and v governs "how large one cell is in this universe."

【Hypothesis; image hypothesis, consistent with *The Alien Cosmology*'s statement that "spacings $v, 1.1v, 1.3v$... distinguish different dimensions"; not rigorously derived in V4】

V4 Chapter 4 gives the 24- fold fractal amplification rule: each layer upward amplifies the spatial scale by 24 times. But the 24- fold rule itself does not specify the absolute scale of the first layer ($n = 0$ layer)—this absolute scale is given by v. In this theory, $v = \ell_P$ (the Planck length), which is experimentally anchored (V4 §1.3 base- space metric).

The hypothetical nature of this proposition lies in: allowing the existence of other "dimensional- state universes" with their own reference spacing $v' \neq \ell_P$. Because v' is different, the grid scales of these universes are completely different from ours; although the two sets of fiber bundles share the layer- number axis n in the base space, their fiber scales differ, so their physical processes (particles, forces) cannot propagate across universes—this provides an image explanation and does not claim to have proven that "other dimensional- state universes are unobservable."

Proposition J.8 (The Necessity of the Five- Step Phase and the Structural Hypothesis of the Five- Band Spacing Spectrum)

Proposition (two- level structure):

First- level necessity (proven): The spiral completes one turn in five steps, producing 5 phase grid points. The proof reproduction of V4 §4.5.2: revolution 5 steps × 5 self- rotation subdivision phases per step $= 25$ phase positions; the start and end points topologically coincide, subtracting 1, giving independent phases $25 - 1 = 24$; 24 is the amplification factor output by discrete sampling. The five- step- per- turn production of 5 phase grid points is proven. (V4 Proposition 4.5.1: five steps per turn, $5 \times 5 - 1 = 24$.)

Second- level structural hypothesis: The intra- layer odd- multiple spacing spectrum also has exactly 5 bands: $1.1v, 1.3v, 1.5v, 1.7v, 1.9v$. It is hypothesized that the five- band odd spectrum and the five- step five- grid- points are two projections of the same intra- layer fivefold discrete structure—one is phase- grid- point counting, the other is spacing- spectrum counting.

Supporting arguments (not proofs):

- Number- theoretic: 5 is prime and coprime to 1,2,3,4, so the 5 bands are independent and non- overlapping;
- Physical: the odd bands are located at the antinodes (half- step positions) between the main nodes, avoiding resonant interference with the even matter states $(2v, 4v, 6v, 8v)$;
- Symmetry: the odd- even dichotomy is isomorphic to the yang- face energy- state/matter- state dichotomy (V3 §1.9).

【Proven: five steps produce 5 phase grid points (V4 §4.5.2); Hypothesis: the intra- layer spacing spectrum also has five bands and its identity with the five- step structure, awaiting rigorous derivation】

The Five Energy States Corresponding to the Five Bands

Odd band	Phase position	Modern physics correspondence	Argumentative status
$1.1v$	First phase state	Photon (no rest mass, edge- midpoint standing wave)	【Proven】V3 §2.4.3; standard physics
$1.3v$	Second phase state	Electromagnetic short- range transmission state (virtual- photon propagator in standard physics)	【Image correspondence】mathematical object, not an observable real particle
$1.5v$	Third phase state	Gravitational wave (Luoshu grid ripple)	【Proven side】LIGO has directly detected; 【correspondence side】V4 §12.2.8 correspondence, image correspondence
$1.7v$	Fourth phase state	Gauge- boson transverse transmission state (gluon, W/Z transverse polarization components)	【Image correspondence】partial correspondence; gauge bosons have mass and require further distinction

$1.9v$	**Fifth phase state (adhesion state)**	**Adhesion state / yin- face starting- point transition state** (starting side of the V3 §1.9 yin- face continuous spectrum)	【Image correspondence】 not a "particle stream" but an adhesion state; the adhesion state is the transition band where things have just separated a little and K is about to split but still adheres, not the void

Reading note: The $1.1v$ photon band is proven in V3 + a standard physical fact; the $1.5v$ gravitational- wave band is "factually proven (LIGO) + correspondence as image language (V4 §12.2.8)"; the $1.3v$, $1.7v$, and $1.9v$ bands are image correspondences. The "virtual photon" of $1.3v$ is a reification of the standard- physics propagator; it does not claim that virtual photons are real particles; $1.9v$ corresponds to the starting side of the yin- face continuous spectrum and is an **adhesion state**—the transition band where things have just separated a little and K is about to split but still adheres; it is not the void, nor an observable particle. **The void is the one- body static- energy block at $\Delta = 0$, where K has not split; the adhesion state is the transition band at Δ just greater than 0, where K has just begun to split but still adheres; the two are not the same thing.** The five bands as "the spacing projection of the intra- layer fivefold discrete structure" are a structural hypothesis; the rigorous particle correspondence between the five bands and the five energy states is an open problem.

Even bands $2v$**–**$8v$ correspond to matter states (fermion standing waves) 【image correspondence】, complementary to the odd energy states, forming the yang- face odd- even dichotomy (V3 §1.9).

【Overall annotation: 1.1v proven; 1.3v/1.7v/1.9v image correspondence; 1.5v factually proven, correspondence as image language】

Part Four: List of Open Problems

1. Rigorous particle correspondence between the intra- layer five- band spacing spectrum and the five energy states (Proposition J.8): it is necessary to establish the quantization conditions of the intra- layer spacing spectrum, rigorously derive why there are exactly 5 bands rather than another number, and give the precise correspondence mapping between each band and Standard Model particles.

2. Consistency of FCA (Axiom F) in standard number theory and a rigorous topological definition of "adhesion" (Proposition J.2): FCA is currently an axiom hypothesis of this theory; V3 has proven through FCA- NSA that it is consistent with ZFC within the non- standard- analysis framework (V3 Proposition 12.3.11.1), but a rigorous mathematical definition of "topological adhesion" (e.g., using homotopy groups, homology groups, etc.) is still needed.

3. The observer's position selecting the facing direction (Proposition J.6): this image needs to be rigorously mathematized—defining the precise counterpart of the observer's position in fiber- bundle geometry, and the mathematical operation of "selecting a direction" (e.g., selecting a cross- section, gauge fixing, etc.).

4. Whether there exists any observable coupling channel between different- v dimensional- state universes (Proposition J.7): although universes with different v are assumed to be mutually isolated, gravity may "leak" through extra dimensions (e.g., the Randall- Sundrum model), and it is necessary to examine whether observable cross- universe effects exist.

5. The identity between the five- band spacing spectrum and the five- step phase grid: whether there exists a structure- preserving map from phase grid points to spacing bands.

6. Rigorous correspondence between the three- ring spherical envelope surface and the V4 Tube projection: the mapping between the envelope surface and the pipeline cross- section.

Part Five: Interface Comparison Table with the V4 Main Text

Proposition	Corresponding V3/V4 section	Argumentative status
J.1 Yang face/yin face/void	V3 §1.9	Proven
J.2 Fractal nesting dissolves divergence	V4 §1, §11; V3 §1.9, §12.3.7; FCA	Argument of this appendix (within FCA)
J.3 Continuity- discreteness isomorphism	V4 §11	Reproduction within the proven framework
J.3bis QM and GR irreconcilability	V4 §11; V3 §12.3	Argument of this appendix (based on V4/V3)
J.1bis Twin primes/Goldbach	V3 §12.3.7–12.3.12	Proven (within FCA)
J.4 Three- ring spherical closure	V4 §1.4, §1.6, §5; V3 Axiom D	Image correspondence

J.5 Luoshu matrix projection	V4 §5	Image correspondence
J.6 Observer direction selection	— (newly added)	Hypothesis
J.7 v and dimension	V4 §4; *The Alien Cosmology*	Hypothesis
J.8 Five-step necessity + five-band hypothesis	V4 §2, §4, §5	Proven (five-step phase grid points) + hypothesis (five-band identity)
J.9 Five-band correspondence table	V3 §2.4.3; V4 §12.2.8; standard physics	Mixed (proven/image correspondence)

Part Six: Declaration

1. This appendix is a supplementary explanation and does not overturn the published conclusions of V3/V4; all numerical verification results of V3/V4 are retained.

2. The parts of this appendix marked "Hypothesis" and "Image correspondence" are correspondences at the image level and are not claimed to be rigorous mathematical theorems.

3. The proof reproductions of the "Proven" items in this appendix are faithful to the original V3/V4 texts; if there is any discrepancy between the reproduction and the original, the original shall prevail.

4. This appendix involves the image background for the intra-layer spacing spectrum $1.1v$–$1.9v$, see reference [X]; this image is only a hypothetical background, and the numerical results and derivations in the main text of V3/V4 do not depend on this source.

One-sentence summary of this appendix: Discrete and continuous are not opposites; they are projections of the same fractal fiber bundle at different layers of the yang face and yin face; the Luoshu nine-grid is the projection of the three-ring orthogonal closed spherical surfaces onto the two-dimensional grid; v is the ruler of dimensional-state universes, and the five odd bands correspond to five energy-state phases—among which the photon band and the gravitational-wave band have proven pillars, the fifth band is the adhesion state (the yin-face starting-point transition state, not the void; the void is one-body with K unsplit, while the adhesion state is where things have just separated a little and K is about to

split but still adheres), and the rest are image correspondences awaiting rigorous derivation; the main ideas of the twin- prime and Goldbach conjectures are precisely that at $\Delta n \to 0$, yang- face discrete points adhere into the same K value in the yin face; the irreconcilability of quantum mechanics and relativity has its root in the fact that each is a projection of the same fractal structure under different limits, and their conflict is not a computational- technical problem but an ontological conflict.

References

[X] Song, Shipeng. Dialogue with Aliens (Wo yu waixingren de duihua) [M]. Hong Kong: Great Wall (Hong Kong) Culture Publishing Company, 2012.